

Comparison of Survival Among Eligible Patients not Enrolled Versus Enrolled in the Collaborative Ocular Melanoma Study (COMS) Randomized Trial of Pre-Enucleation Radiation of Large Choroidal Melanoma

Marta M. Gilson, Marie Diener-West & Barbara S. Hawkins

To cite this article: Marta M. Gilson, Marie Diener-West & Barbara S. Hawkins (2007) Comparison of Survival Among Eligible Patients not Enrolled Versus Enrolled in the Collaborative Ocular Melanoma Study (COMS) Randomized Trial of Pre-Enucleation Radiation of Large Choroidal Melanoma, *Ophthalmic Epidemiology*, 14:4, 251-257, DOI: [10.1080/01658100701473275](https://doi.org/10.1080/01658100701473275)

To link to this article: <https://doi.org/10.1080/01658100701473275>



Published online: 08 Jul 2009.



Submit your article to this journal [↗](#)



Article views: 148



View related articles [↗](#)



Citing articles: 2 View citing articles [↗](#)

Comparison of Survival Among Eligible Patients not Enrolled Versus Enrolled in the Collaborative Ocular Melanoma Study (COMS) Randomized Trial of Pre-Enucleation Radiation of Large Choroidal Melanoma

Marta M. Gilson,¹ Marie Diener-West,^{2,3,4} and Barbara S. Hawkins^{2,3}

¹Department of Surgery, School of Medicine, Johns Hopkins University

²Department of Ophthalmology, School of Medicine, Johns Hopkins University

³Department of Epidemiology, School of Public Health, Johns Hopkins University

⁴Department of Biostatistics, School of Public Health, Johns Hopkins University

ABSTRACT

Purpose: To compare survival between patients enrolled in the Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation therapy (PERT) for large choroidal melanoma and eligible patients who did not enroll. **Methods:** COMS clinical center personnel prospectively reported to the COMS Coordinating Center all patients with choroidal melanoma examined between November 1986 and December 1994. Deaths of enrolled patients were reported prospectively by clinical center personnel. In a COMS ancillary study, we retrospectively searched medical records of participating clinical centers, the Social Security Death Index, and the National Death Index to determine vital status of eligible patients not enrolled. Cox proportional hazards analysis was used to compare survival within 10 years of baseline reporting and before July 31, 2000, of enrolled patients versus eligible patients not enrolled. **Results:** Clinical centers that received local institutional review board approval to participate in this ancillary study prospectively reported on 129 of 299 eligible patients not enrolled in the COMS PERT trial. The baseline characteristics of the 129 patients included in this ancillary study were similar to those of the 170 patients not included; 73 patients were reported as deceased. Previously identified prognostic covariates, i.e., age and longest tumor diameter, were confirmed to predict survival in both enrolled patients and eligible patients not enrolled; trial enrollment was not predictive. After adjusting for prognostic covariates and stratifying by clinical center, the estimated hazard ratio (enrolled vs. not-enrolled) was 1.12 (95% confidence interval: 0.83 to 1.51). **Conclusions:** The results of the COMS PERT trial should be generalizable to all patients with choroidal melanoma meeting the eligibility criteria for that trial. While the methods we used may not be generalizable to all clinical trials because of unique features of the COMS, other researchers may be able to use similar methods to determine the generalizability of their trial results.

Received 9 February 2006, Accepted 23 May 2007

Dr. Hawkins neither selected peer reviewers nor participated in any other aspect of the peer review process for this manuscript.

Keywords: Clinical trials, Generalizability, National Death Index, Participation, Social Security Death Index

Correspondence to:

Marta M. Gilson

550 North Broadway, 9th Floor

Baltimore, MD 21205, USA

email: mmgilson@jhmi.edu

This manuscript is based on a PhD dissertation available at <http://www.lib.umi.com/dissertations/fullcit/3197198> (Accessed March 29, 2007).

Portions of this manuscript were presented to the Society for Clinical Trials, May 22, 2006, Orlando Florida (published abstract: *Clinical Trials* 2006;3:224).

This investigation was performed according to the guidelines of the Declaration of Helsinki.

INTRODUCTION

A common criticism of randomized clinical trials is that eligibility and exclusion criteria, the choices of treatments to be assigned in the trial, and the source of the patients limit the ability of physicians to apply the findings in clinical practice. To address this criticism, a number of comparisons have been made of the outcomes of patients enrolled in clinical trials to those not enrolled.^{1–3} However, such comparisons have been challenging because the universe of patients not enrolled in a given clinical trial usually is large and cannot be completely identified and described.^{4–6} Because the Collaborative Ocular Melanoma Study (COMS) Group prospectively identified all eligible patients who did not enroll in the COMS,^{7–10} we had a unique opportunity to address this issue.

In the United States, uveal melanoma is the most common primary malignant intraocular tumor in adults;¹¹ it frequently metastasizes, primarily to the liver with high mortality.^{12,13} Unfortunately, choroidal melanoma is often asymptomatic until the tumor affects vision. The most important prognostic factors for survival reported by the COMS Group and others include patient age and tumor size.^{14–17}

In the COMS clinical trial of interest in this report, participants with “large” choroidal melanoma⁹ were randomized to either enucleation alone (removal of the eye) or pre-enucleation radiation therapy (PERT). The primary outcome was all-cause mortality; 43 clinical centers in the United States and Canada participated in this clinical trial between November 1986 and July 2000. The feasibility of enrolling a sufficient number of patients in the COMS was monitored during the accrual period by reports of all patients diagnosed with choroidal melanoma from each clinical center. Complete baseline data were collected for trial participants; a reduced set of baseline data was collected for eligible patients who did not enroll.

The first patient enrolled in the COMS PERT trial in November 1986; accrual ended on December 15, 1994. All clinical follow-up ended on July 31, 2000. Mortality findings for the PERT trial participants have been published;^{14,15} no statistically or clinically significant difference in mortality by treatment was found.

To evaluate the generalizability of the COMS PERT trial results, we conducted an ancillary study to compare the survival of patients enrolled in the PERT trial with survival of patients who were eligible for the trial but did not enroll.

PATIENTS AND METHODS

The COMS Executive Committee approved the current study as an ancillary study to the COMS PERT trial and granted access to the baseline data for all trial participants and for eligible but not enrolled patients and to vital status information for trial participants. Baseline information collected about these eligible but not enrolled patients included age, gender, race, education, and longest basal diameter of the tumor. The reason for not enrolling and the proposed management strategy of the choroidal melanoma were reported for most of the eligible patients who

did not enroll in the COMS PERT trial. Mortality rates in the PERT trial through July 31, 2000, and within 10 years after enrollment have been published previously.¹⁵

This study was approved by the Johns Hopkins School of Medicine institutional review board (IRB). Participation and local IRB approval were requested from each COMS clinical center, beginning in 2000. After approval was obtained, vital status of the eligible patients who did not enroll was determined from searches of local medical records at each clinical center, searches of the Social Security Death Index (SSDI) (www.ancestry.com), and a search of the National Death Index (NDI).

In order to facilitate a central search of the SSDI and the NDI, local clinical center personnel provided personal identifying information for any patient who was known to be deceased based on the medical records of the clinical center and for any patient who was presumed to be alive but whose most recent contact with the clinical center was more than 6 months in the past. Whenever personnel at the COMS clinical center had recent contact (within 6 months) with an eligible patient who did not enroll, and thus the patient was known to be alive at that time, no personal identifying information was provided.

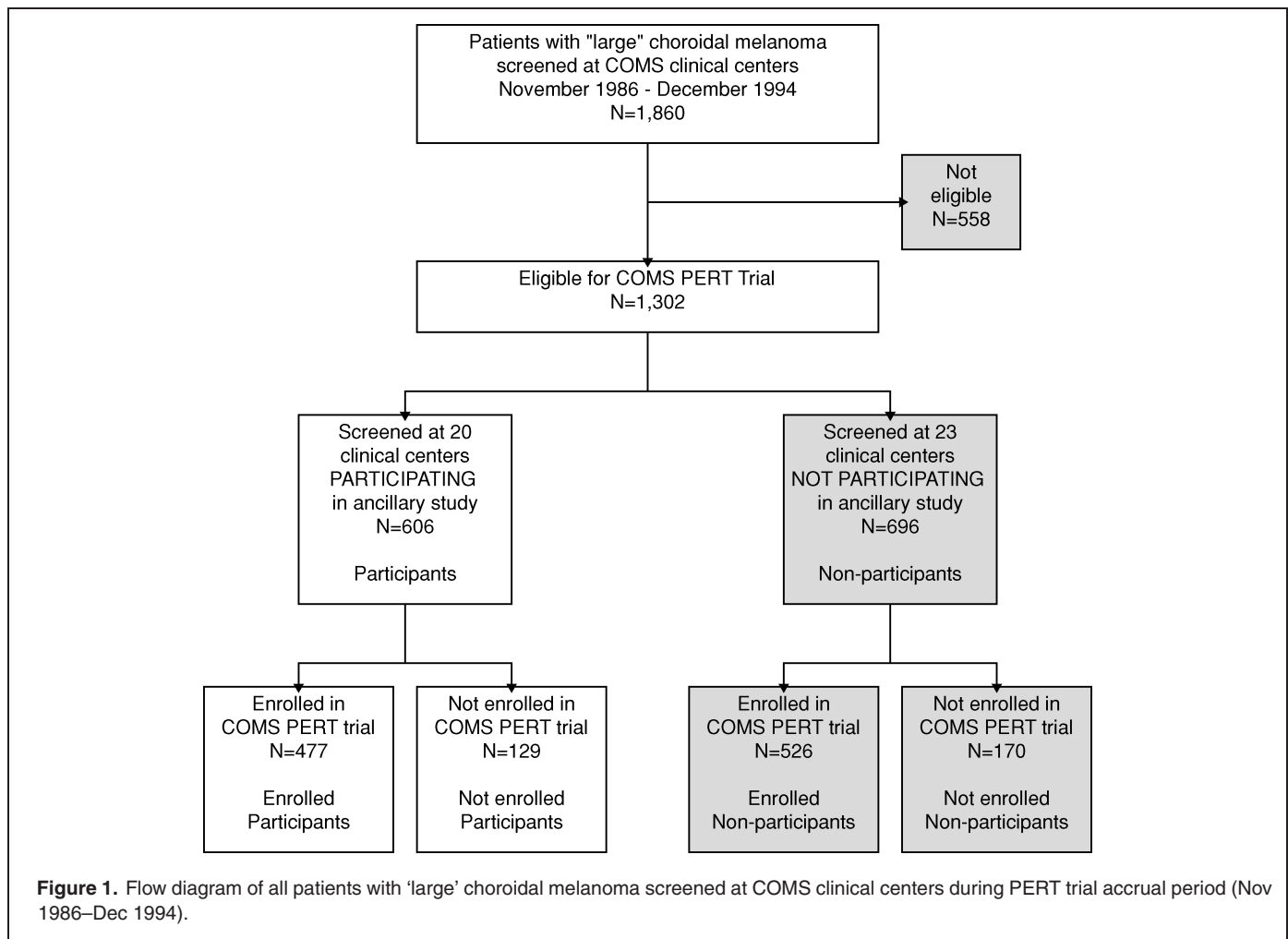
To satisfy local IRB concerns, some clinical centers participated without providing personal identifying information. In such cases, local clinical center personnel searched the SSDI and provided only the vital status search results (date of death if found, “no match” if not found). Thus, patients from these centers were not included in the central search of the NDI.

Comparisons of distributions of categorical baseline variables between enrolled and eligible patients not enrolled used the chi-squared statistic. Student’s *t*-test was used to compare means of continuous baseline variables. Kappa statistics were calculated to compare vital status results from the SSDI and NDI. Cox proportional hazards analysis was used to compare survival within 10 years of baseline reporting and before July 31, 2000 of enrolled patients (published outcome¹⁵) versus eligible patients not enrolled. Statistical analyses were performed using commercially available software (SAS release 8.2; SAS Institute Inc., Cary, NC); figures were created using Ploticus free ware (<http://ploticus.sourceforge.net>).

RESULTS

Of the 43 COMS PERT trial clinical centers, 2 Canadian centers were excluded because the NDI and SSDI only include deaths in the United States or decedents with U.S. social security numbers. Principal investigators at 12 additional COMS centers opted not to participate; 1 center closed COMS operations before IRB approval could be obtained; the principal investigator and personnel willing to participate at 4 clinical centers were prevented by local IRB disapproval of the study; 4 centers enrolled all patients eligible for the COMS PERT trial. Thus, 20 clinical centers with IRB approval participated in this ancillary study. The decision to participate was made at the clinical center level and not at the patient level.

Figure 1 shows the distribution of 1,860 patients with “large” choroidal melanoma screened for the PERT trial at COMS



clinical centers; of these, 1,302 were eligible for enrollment. Of the 1,302 eligible patients, 606 were screened at the 20 clinical centers participating in this study; 477 (79%) were enrolled in the COMS PERT trial and 129 (21%) were eligible but not enrolled. Of the 696 non-participants (screened at 23 clinical centers not participating), 526 (76%) were enrolled in the COMS PERT trial and 170 (24%) were eligible but not enrolled.

Comparison of Baseline Variables

The COMS Group published a comparison of baseline characteristics for all individuals screened for the COMS PERT trial.⁹ Table 1 summarizes baseline information by the four groups defined by participation in the ancillary study and PERT trial enrollment status. All groups were similar with respect to baseline age, gender, and race of patients. There was a slight imbalance in education level and tumor size among the four groups, but among the participants in the ancillary study these variables were not statistically different between patients enrolled in the COMS PERT trial and those eligible but not enrolled.

Vital Status for Eligible not Enrolled Ancillary Study Participants

Personnel at each participating clinical center searched the local medical records to determine vital status of the eligible patients not enrolled. One patient had no data available at the local clinic; 27 (21%) of the remaining 128 patients were reported as deceased based on local records. Dates of death ranged from September 1988 to April 2001.

Of the 128 patients, 45 were screened at clinical centers that searched the online Social Security Death Index (SSDI) and reported the results of the search on a preprinted form without providing any personal identifying information; 83 were screened at clinical centers with IRB permission to provide personal identifiers. Of the latter 83 patients, 7 were known to be alive and had contact with someone at the clinical center within the previous six months; personal identifiers were not requested for them. Personal identifying information for the remaining 76 patients was provided for central searches of the NDI and the SSDI. The number of SSDI matches (deceased individuals) was 70 (55%), leaving 58 without matches (presumed living). Dates of death for the 70 matches ranged from May 1988 to October 2002.

Table 1. Baseline characteristics of eligible patients enrolled and not enrolled in the COMS randomized trial of pre-enucleation radiation therapy (PERT) by clinical center participation in ancillary study

	Participants in ancillary study		Non-participants in ancillary study	
	Enrolled (n = 477)	Eligible not Enrolled (n = 129)	Enrolled (n = 526)	Eligible not Enrolled (n = 170)
	Mean (Standard deviation)			
Age (yrs)	59.5 (13.8)	59.3 (16.1)	60.6 (13.6)	61.6 (14.7)
Longest Basal Diameter (mm)	17.0 (3.0)	16.4 (3.4)	17.5 (2.9)	16.9 (2.9)
	N (%)			
Women	205 (43)	54 (42)	220 (42)	86 (52)
Caucasian *	461 (97)	125 (97)	509 (97)	164 (96)
> High School Education	173 (37)	36 (46)	160 (31)	42 (39)
Reason for not enrolling in PERT Trial:				
Treatment preference	—	59 (46)	—	77 (45)
Refused randomization	—	34 (26)	—	40 (24)
Other	—	36 (28)	—	53 (31)
Treatment				
Enucleation alone	242 (51)	88 (68)	264 (50)	121 (71)
PERT	235 (49)	15 (12)	262 (50)	16 (9)
Other	—	26 (20)	—	33 (19)

*Self-reported.

The NDI was searched for deaths during the years 1986 through 2000 for the 76 patients with personal identifiers provided. The NDI search yielded 56 matches; however, 18 were determined not to be true matches based on all available patient information (for example major discrepancies in social security number, date of birth or other information) and were presumed to be alive for data analysis purposes. The final number of NDI matches (deceased individuals) was 38 (50%). Dates of death for the 38 matches ranged from September 1988 to October 2000.

Vital status indicators from all search methods were combined to create a single vital status variable, summarized in Table 2. Patients were reported as deceased by any method, were counted as deceased. Whenever there were multiple dates of death, the earlier date was used. Of the 128 patients with survival data, 73 (57%) were reported as deceased by at least one method.

Comparison of Mortality Information from SSDI and NDI

Central searches of both the National Death Index (NDI) and the Social Security Death Index (SSDI) were performed for the 76 patients for whom personal identifying information was

provided. The NDI was searched for deaths only through the year 2000, to conform to the mortality follow-up period for the COMS PERT trial participants; thus, post-2000 deaths discovered through the SSDI were re-classified as alive for this comparison. Thirty-six (47%) patients were judged to be alive after both searches, 36 decedents were identified by both searches, 2 (3%) decedents were identified by SSDI but not NDI, and 2 (3%) decedents were identified by NDI but not SSDI. The Kappa statistic¹⁸ for the agreement between NDI and SSDI was 0.89, 95% confidence interval (CI) (0.79 to 1.00), suggesting excellent agreement.

Comparison of Overall Survival

The cutoff date of July 31, 2000, which marked the end of clinical follow-up and data collection in the PERT trial, was used for the comparison of survival among patients enrolled versus eligible patients not enrolled. To correspond to published data,¹⁵ survival of enrolled patients was censored 10 years after randomization. A comparable 10-year censoring date was computed based on the “start date” chosen from baseline dates for each eligible patient not enrolled. Overall there were 59 deaths among eligible patients not enrolled, identified by at least one method, that occurred before July 31, 2000, and within 10 years of the start date.

Figure 2 shows the survival estimates, with 95% confidence intervals of the ancillary study participants (128 eligible but not enrolled patients and corresponding 477 patients enrolled in the COMS PERT trial at participating centers). Cox proportional hazards analysis was used to investigate the relationship between survival and enrollment in the COMS PERT trial. The proportionality assumption was tested¹⁹ and was not rejected ($p = 0.71$). The unadjusted hazard ratio for death by enrollment status was 1.30 (95% CI: 0.98 to 1.72) suggesting a possible small increase in the hazard of death for those enrolled compared

Table 2. Vital status of eligible patients not enrolled in the COMS PERT trial, by source of information

Source of vital status information	Eligible not Enrolled		Total
	Dead N (%)	“Alive” N (%)	
Medical Records	27 (21)	101 (79)	128
SSDI	70 (55)	58 (45)	128
NDI	38 (50)	38 (50)	76
Any Source	73 (57)	55 (43)	128

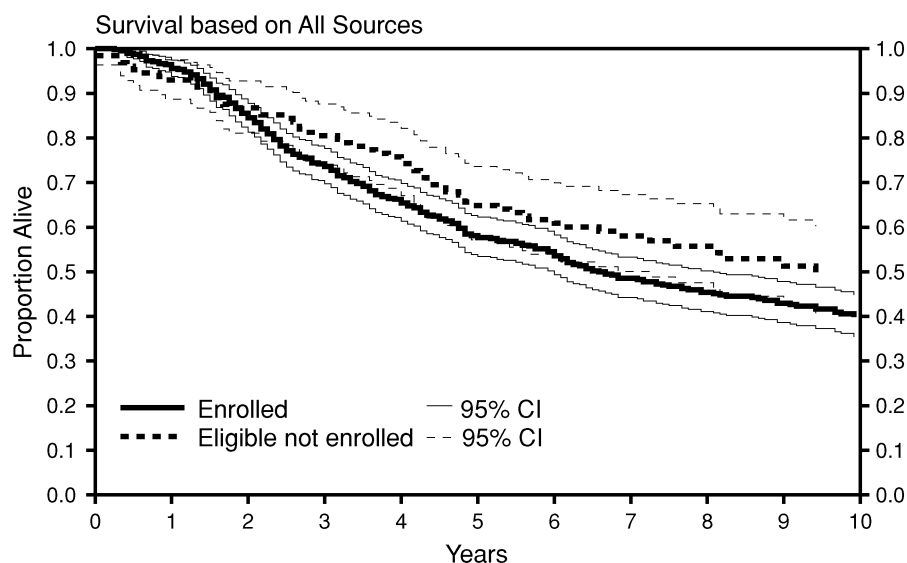


Figure 2. Cox proportional hazards survival curves and 95% confidence intervals for eligible patients enrolled ($n = 477$) and not enrolled ($n = 128$). Solid lines correspond to patients enrolled in COMS PERT trial. Dotted lines correspond to eligible patients not enrolled in COMS PERT trial. Overlapping 95% confidence intervals reflect absence of statistically significant differences between the groups.

to eligible patients not enrolled. Two covariates (patient age and tumor longest basal diameter) were included in a second Cox proportional hazards model along with the dichotomous variable representing enrollment in the COMS PERT trial. Linearity assumptions for these continuous variables were checked and appeared reasonable. The hazard ratio for death by enrollment status was 1.24 (95% CI: 0.93 to 1.64).

After stratifying by clinical center and adjusting for age and longest tumor basal diameter, the hazard ratio for enrolled participants declined to 1.12 (95% CI: 0.83 to 1.51). In this final model, older age and larger basal tumor diameter remained statistically significantly associated with mortality. Trial enrollment was not a statistically significant independent predictor of mortality in either unadjusted or adjusted models. Table 3 shows the hazard ratios and associated 95% confidence intervals (CIs) for the different proportional hazard models.

DISCUSSION

Patients who enrolled in the COMS PERT trial were very similar to the eligible patients who did not enroll; the treatments received were standard treatments and have been shown not to differ statistically or clinically in effect on overall mortality. The hazard of death among eligible patients who did not enroll was similar to that of trial participants, particularly after stratifying by clinical center and adjusting for known prognostic factors (age and tumor size). These findings support the statement of the COMS Group that “findings from this clinical trial apply to all patients who have large choroidal melanoma and satisfy COMS eligibility criteria.”⁹ As reported elsewhere,⁹ patients who were judged ineligible for the PERT trial already had metastatic melanoma for which local treatment would not be effective or other characteristics that placed them at increased risk of premature death.

Table 3. Unadjusted and adjusted estimates from cox proportional hazards models

Characteristic	Unadjusted HR (95% CI)	Adjusted* HR (95% CI)	Adjusted* Stratified** HR (95% CI)
Trial enrollment	1.30 (0.98–1.72)	1.24 (0.93–1.64)	1.12 (0.83–1.51)
Age (years)	1.03 (1.02–1.04)	1.02 (1.02–1.03)	1.02 (1.02–1.03)
LBD (mm)	1.15 (1.11–1.19)	1.13 (1.09–1.17)	1.14 (1.10–1.19)

LBD = longest basal diameter of the tumor; HR = Hazard ratio of death; CI = Confidence interval.

* = Adjusted analysis, all three variables in model.

** = Stratified analysis on clinical center.

Strengths and Limitations

The primary strength of this ancillary study is that the COMS Coordinating Center maintained a prospective record of each patient diagnosed with choroidal melanoma and screened for the COMS at participating centers. It was possible, using these data, to identify a group of eligible non-enrolled patients who were diagnosed and evaluated at the same centers concurrently with patients who were enrolled in the COMS PERT trial. It would have required a much greater effort and resulted in a much lower yield if it had been necessary to identify this group of patients retrospectively from local clinical center records only.

It is of interest that even in 1986, when the COMS began, IRBs expressed concern when a proposal to conduct an observational study of these patients was considered. With current HIPAA regulations and other privacy issues, a retrospective study of this type may be impossible.

A limitation of this ancillary study was the incomplete participation of the clinical centers; reasons for non-participation included lack of funding or personnel, and other issues. There also was variability regarding local IRB requirements and decisions, which resulted in some clinical centers being denied permission to participate in this ancillary study and differences in release of personal identifying information for local versus central ascertainment of vital status. Variation in IRB decisions in multicenter studies has been documented previously as problematic.²⁰

However, apart from less precise estimates of survival rates, we have no reason to believe that the findings were biased due to lack of participation, particularly after comparison of baseline patient characteristics between participating and non-participating centers (Table 1) and comparison of outcomes of PERT trial enrollees between participating and non-participating centers (data not shown).

Unlike the COMS PERT trial enrollees who were followed prospectively at scheduled clinical examinations for 5 to 13 years, vital status information for all eligible patients who did not enroll was obtained retrospectively using multiple sources of information. Therefore, it is possible that our estimate of the hazard ratio is biased. Medical records, the National Death Index (NDI) and the Social Security Death Index (SSDI) are each imperfect sources of current vital status information and do not always agree (Table 2).

Despite some disagreements among sources of vital status information, a major strength of the study was the ability to collect vital status data with minimal patient contact, relatively minimal effort by the clinical center personnel and the ancillary study personnel, minimal cost, and availability of two standard national databases (NDI and SSDI) previously documented to be reliable and to have good sensitivity and specificity.^{21,22} Collecting data from multiple sources provided the ability to evaluate the reliability and accuracy of the data obtained. As the two databases produced results with excellent agreement, it is likely that classification of decedents was reliable.

The assumption regarding individuals not found in either database as "living" is more subject to error. But it is unlikely

that we missed enough deaths in the group of eligible patients not enrolled to change the estimate of the hazard ratio substantially.

Generalizability

The COMS experience was unique in three respects. First, choroidal melanoma is a rare condition; typically cases are referred to centers with special expertise in its diagnosis and treatment. Thus, a large proportion of cases diagnosed in the United States and Canada during the accrual period are believed to have been reported by COMS centers. Second, although not implemented for this purpose, prospective reporting of all cases facilitated the conduct of this ancillary study, as discussed above.

Third, all cause mortality was the primary outcome of the COMS. In the United States there are publicly available databases that can be used to determine vital status of individuals if researchers have the necessary linkage variables (i.e. social security number, name, date of birth, etc.). However, outside the United States many countries have national databases containing clinical health information that potentially could be used to determine many types of clinical outcomes^{23–26} for eligible individuals not enrolled in randomized clinical trials. In such settings, similar methods that utilize prospective identification of eligible but not enrolled individuals and recording of identifying information to allow linkage to databases with outcome data, can be applied to other ocular and non-ocular conditions.

ACKNOWLEDGMENTS

The authors received support during the conduct of this study from the National Institutes of Health, US Department of Health and Human Services, Bethesda, Maryland, through a training grant T32 EY07127 and a cooperative agreement U10 EY06287.

This research could not have been completed without the help of the COMS Principal Investigators and Clinic Coordinators, listed by city. (Numbers in parentheses indicate the number of centers that participated in this ancillary study.)

Atlanta, GA (2): Paul Sternberg, MD; James H. Frank, MD; Jayne M. Brown, BA

Baltimore, MD: Andrew P. Schachat, MD; Warren T. Doll, Jr., COA

Boston, MA: Clement Trempe, MD; Zena F. Nies

Chicago, IL (2): Lee M. Jampol, MD; Norman P. Blair, MD; Lori Kaminski, BSN, MS, RN

Cleveland, OH: Z. Nicholas Zakov, MD; Stephanie A. Schura, COT

Columbus, OH: Frederick H. Davidorf, MD; Cynthia S. Taylor, BS

Dallas, TX: Dwain Fuller, MD; Sally A. Arceneaux, COST, COA

Denver, CO: Kenneth R. Hovland, MD; Suzette Compton, RN

Houston, TX: Richard S. Ruiz, MD; Dalia Vargas

Iowa City, IA: Thomas A. Weingeist, MD, PhD; Connie Fountain, COT, CCRC

Los Angeles, CA: Robert E. Engstrom, Jr, MD; Robert D. Almanzor

Madison, WI: Suresh R. Chandra, MD; Kathryn F. Burke, BA
 New York, NY: Paul T. Finger, MD; Kimberly J. Chin, OD
 Oklahoma City, OK: Reagan H. Bradford, Jr, MD; Lisa M. Ogilbee, COA, BS
 Portland, OR: David J. Wilson, MD; Shirley Ira, COT
 Royal Oak, MI: Antonio Capone, Jr, MD; Kristi Cumming, RN, MSN
 St. Louis, MO: J. William Harbour, MD; Mary V. Gaines, BS
 Seattle, WA: Craig G. Wells, MD; Betty Lawrence, BS, CCRP

REFERENCES

- Antman K, Amato D, Wood W, Corson J, Suit H, Proppe K, Carey R, Greenberger J, Wilson R, Frei E 3rd. Selection bias in clinical trials. *J Clin Oncol* 1985 Aug;3(8):1142–1147.
- Bahit MC, Cannon CP, Antman EM, Murphy SA, Gibson CM, McCabe CH, Braunwald E. Direct comparison of characteristics, treatment, and outcomes of patients enrolled versus patients not enrolled in a clinical trial at centers participating in the TIMI 9 Trial and TIMI 9 Registry. *Am Heart J* 2003 Jan;145(1):109–117.
- Smith P, Arnesen H. Mortality in non-consenters in a post-myocardial infarction trial. *J Intern Med* 1990 Sep;228(3):253–256.
- Britton A, McKee M, Black N, McPherson K, Sanderson C, Bain C. Threats to applicability of randomised trials: Exclusions and selective participation. *J Health Serv Res Policy* 1999 Apr;4(2):112–121.
- Feuer EJ, Frey CM, Brawley OW, Nayfield SG, Cunningham JB, Geller NL, Bosl GJ, Kramer BS. After a treatment breakthrough: A comparison of trial and population-based data for advanced testicular cancer. *J Clin Oncol* 1994 Feb;12(2):368–377.
- Vist GE, Hagen KB, Devereaux PJ, Bryant D, Kristoffersen DT, Oxman AD. Outcomes of patients who participate in randomised controlled trials compared to similar patients receiving similar interventions who do not participate. *Cochrane Database of Methodology Reviews* 2004; Issue 4. Art. No.: MR000009.
- Collaborative Ocular Melanoma Study Group: Design and methods of a clinical trial for a rare condition: The Collaborative Ocular Melanoma Study. COMS Report No. 3. *Control Clin Trials* 1993 Oct;14(5):362–391.
- Collaborative Ocular Melanoma Study Group. Sociodemographic and clinical predictors of participation in two randomized trials: Findings from the Collaborative Ocular Melanoma Study. COMS Report No. 7. *Control Clin Trials* 2001 Oct;22(5):526–537.
- Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma I: Characteristics of patients enrolled and not enrolled. COMS Report No. 9. *Am J Ophthalmol* 1998 Jun;125(6):767–778.
- Collaborative Ocular Melanoma Study Group: The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma. II. Characteristics of patients enrolled and not enrolled. COMS Report No. 17. *Arch Ophthalmol* 2001 Jul;119(7):951–965.
- Chang AE, Karnell LH, Menck HR. The National Cancer Data Base report on cutaneous and noncutaneous melanoma: A summary of 84,836 cases from the past decade. The American College of Surgeons Commission on Cancer and the American Cancer Society. *Cancer* 1998 Oct 15;83(8):1664–1678.
- Patel JK, Didolkar MS, Pickren JW, Moore RH. Metastatic pattern of malignant melanoma. A study of 216 autopsy cases. *Am J Surg* 1978 Jun;135(6):807–810.
- Collaborative Ocular Melanoma Study Group. Assessment of metastatic disease status at death in 435 patients with large choroidal melanoma in the Collaborative Ocular Melanoma Study (COMS): COMS Report No. 15. *Arch Ophthalmol*. 2001 May;119(5):670–676.
- Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma II: Initial mortality findings. COMS Report No. 10. *Am J Ophthalmol* 1998 Jun;125(6):779–796.
- Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma: IV. Ten-year mortality findings and prognostic factors. COMS Report Number 24. *Am J Ophthalmol* 2004 Dec;138(6):936–951.
- Collaborative Ocular Melanoma Study Group: The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, III: Initial mortality findings. COMS Report No. 18. *Arch Ophthalmol* 2001 Jul;119(7):969–982.
- Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: V. Twelve-year mortality rates and prognostic factors: COMS Report No. 28. *Arch Ophthalmol* 2006 Dec;124(12):1684–1693.
- Landis JR, Koch GG. The measurement of observer agreement for categorical data. *Biometrics* 1977 Mar;33(1):159–174.
- Collett D. *Modelling Survival Data in Medical Research* London: Chapman and Hall, 1994; pp. 192–195.
- McWilliams R, Hoover-Fong J, Hamosh A, Beck S, Beaty T, Cutting G. Problematic variation in local institutional review of a multicenter genetic epidemiology study. *JAMA* 2003 Jul 16;290(3):360–366.
- Davis KB, Fisher L, Gillespie MJ, Pettinger M. A test of the National Death Index using the Coronary Artery Surgery Study (CASS). *Control Clin Trials*. 1985 Sep;6(3):179–191.
- Sesso HD, Gaziano JM, VanDenburgh M, Hennekens CH, Glynn RJ, Buring JE. Comparison of baseline characteristics and mortality experience of participants and nonparticipants in a randomized clinical trial: The Physicians' Health Study. *Control Clin Trials* 2002 Dec;23(6):686–702.
- Semmens JB, Lawrence-Brown MM, Fletcher DR, Rouse IL, Holman CD. The Quality of Surgical Care Project: a model to evaluate surgical outcomes in Western Australia using population-based record linkage. *Aust N Z J Surg* 1998 Jun;68(6):397–403.
- Olsen J, Melbye M, Olsen SF, Sorensen TI, Aaby P, Andersen AM, Taxbol D, Hansen KD, Juhl M, Schow TB, Sorensen HT, Andresen J, Mortensen EL, Olesen AW, Sondergaard C. The Danish National Birth Cohort-its background, structure and aim. *Scand J Public Health* 2001 Dec;29(4):300–307.
- Sennerby U, Farahmand B, Ahlbom A, Ljunghall S, Michaelsson K. Cardiovascular diseases and future risk of hip fracture in women. *Osteoporos Int* 2007 May 10.
- Tang ST, Chen ML, Huang EW, Koong SL, Lin GL, Hsiao SC. Hospice utilization in Taiwan by cancer patients who died between 2000 and 2004. *J Pain Symptom Manage* 2007 Apr;33(4):446–453.